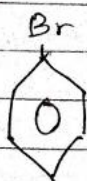


chapter-2

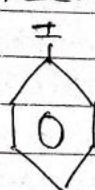
Haloarene (Chlorobenzene).



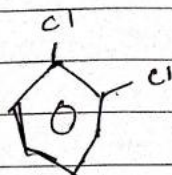
Chlorobenzene



Bromobenzene

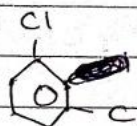


Iodobenzene.



O-dichlorobenzene.

1,2-dichlorobenzene.



M-dichlorobenzene.

1,3-dichlorobenzene.

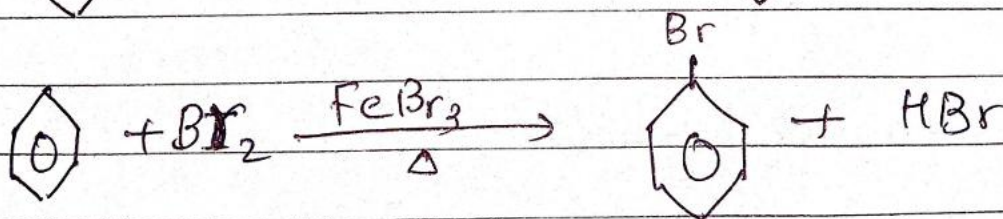
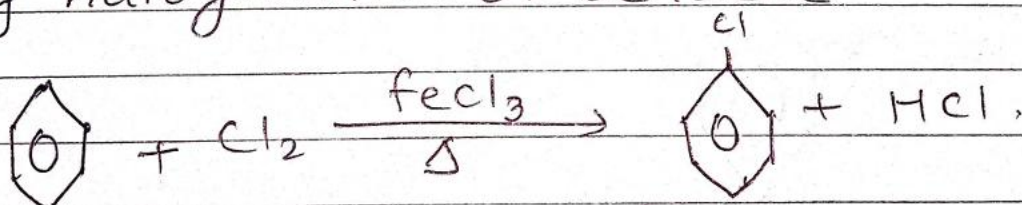


p-dichlorobenzene.

1,4-dichlorobenzene.

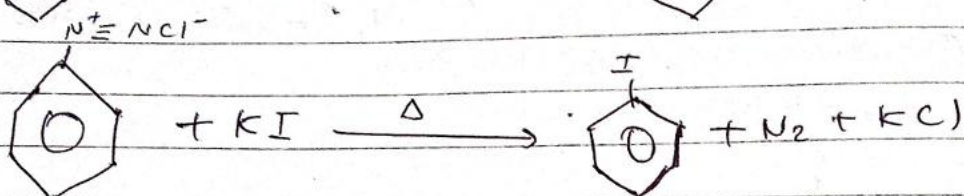
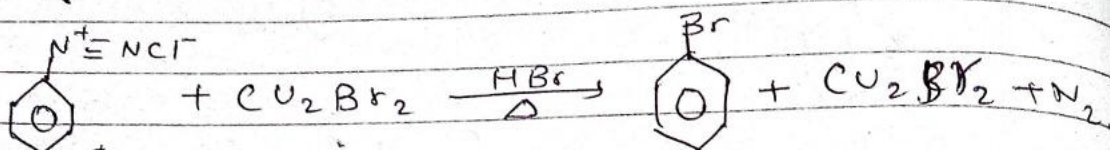
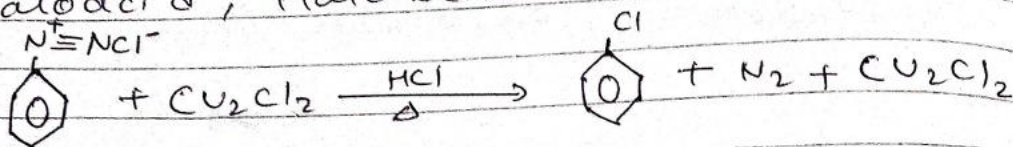
General preparation of ~~benzene~~ haloarene (halobenzene).

① By halogenation of benzene.

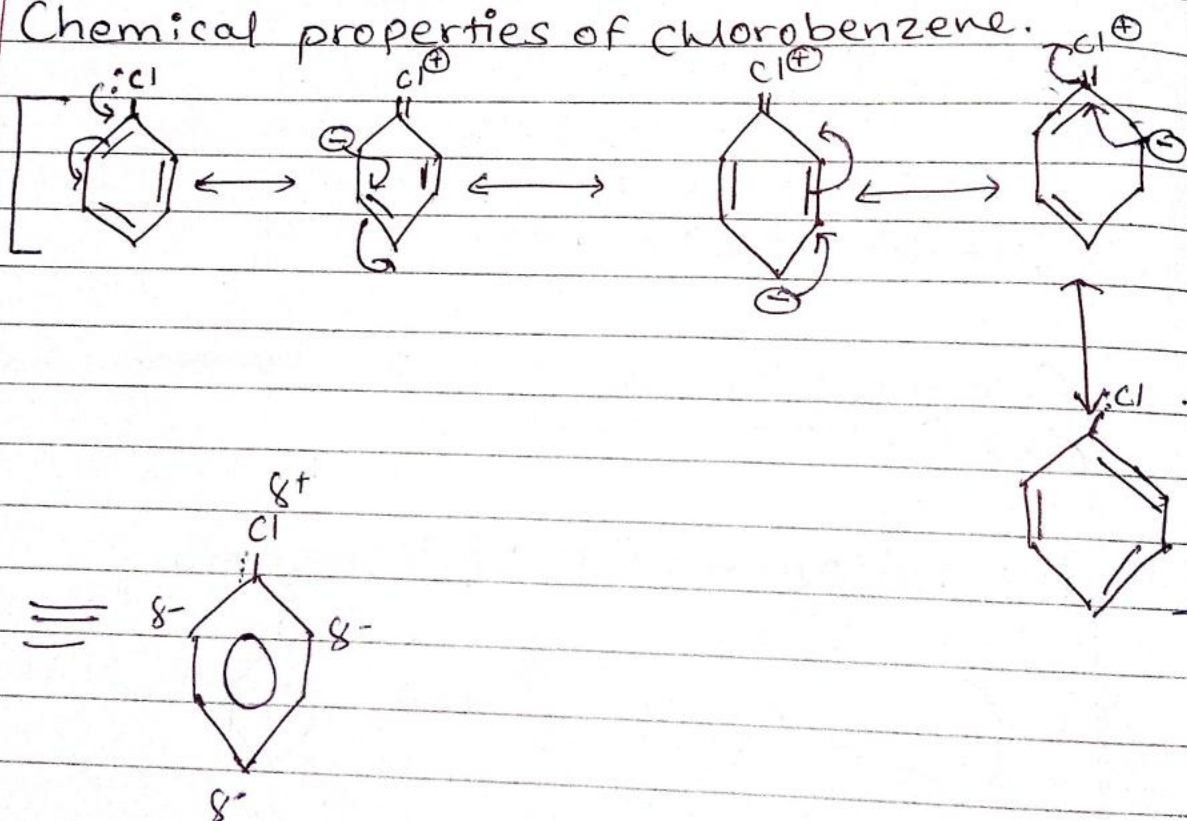


① from benzene diazonium chloride.
(Sandmeyer's rxn).

⇒ Benzene diazonium chloride when heated with cuprous halide in presence of haloacid, Halo benzene is formed.



* Chemical properties of chlorobenzene.



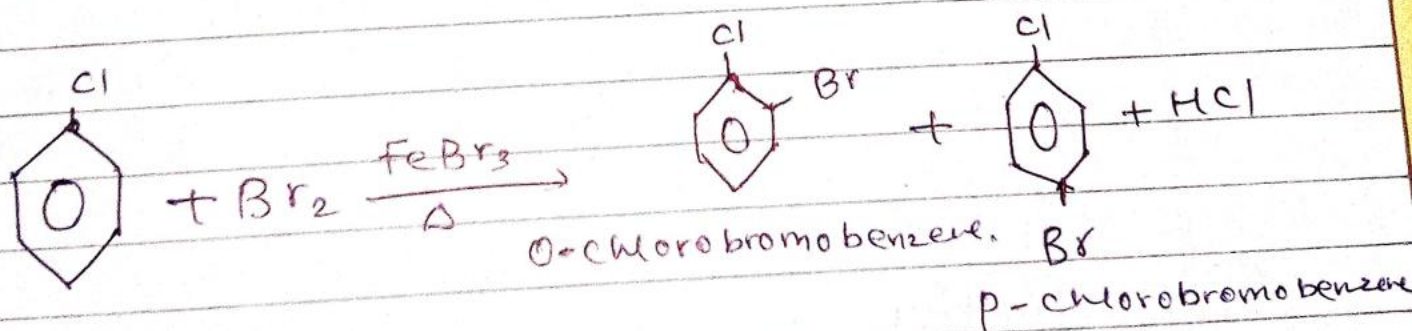
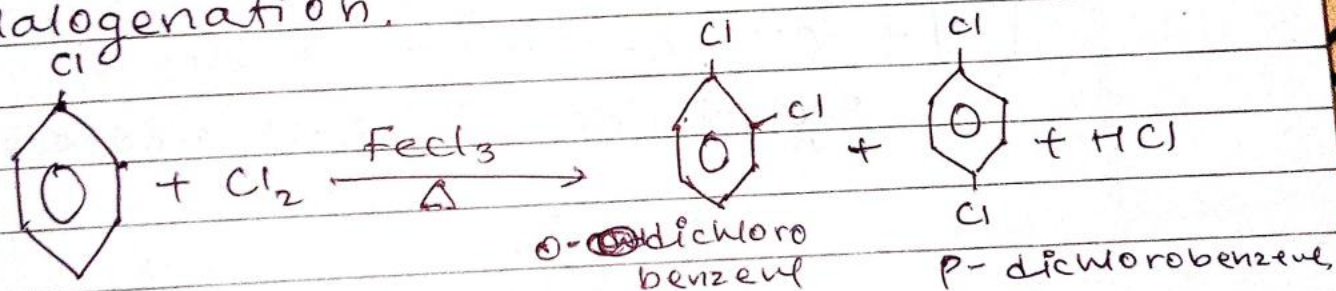
Resonance hybrid.

→ Chlorobenzene doesn't easily undergo nucleophilic substitution rxn because resonance hybrid of chlorobenzene shows partial double bond between carbon and chlorine which is stronger than single bond and doesn't dissociate easily.

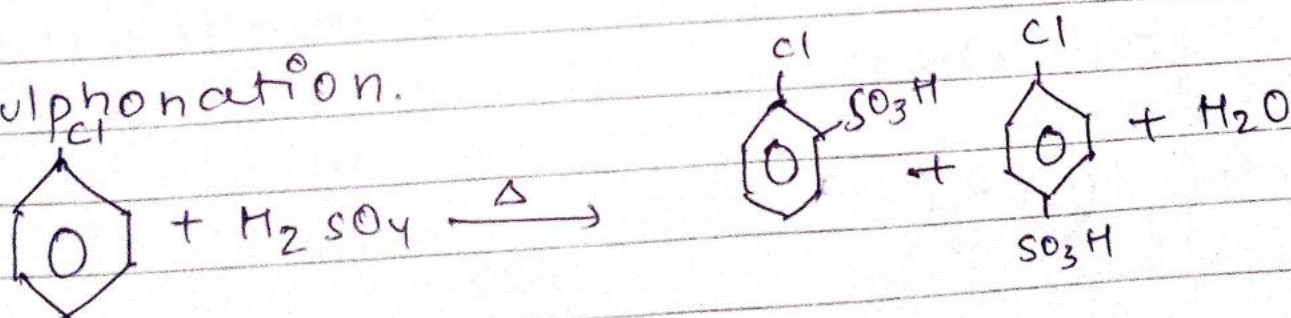
(#) Aromatic electrophilic substitution.

→ Resonance hybrid of chlorobenzene shows partial negative charge at ortho and para position. Since electrophile prefer -ve charge, it attacks to negatively charged op position. So, in case of chlorobenzene (halobenzene), electrophilic substitution occurs at o and p position.

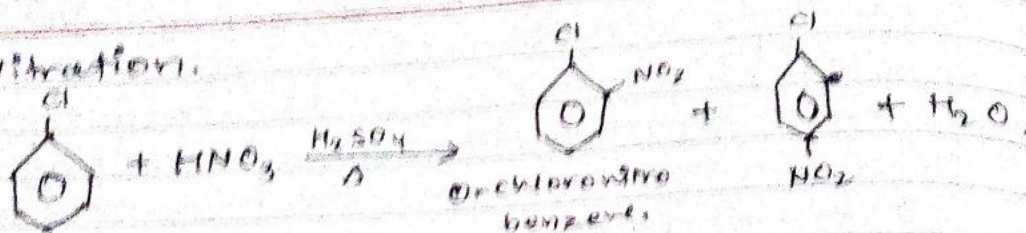
a) Halogenation.



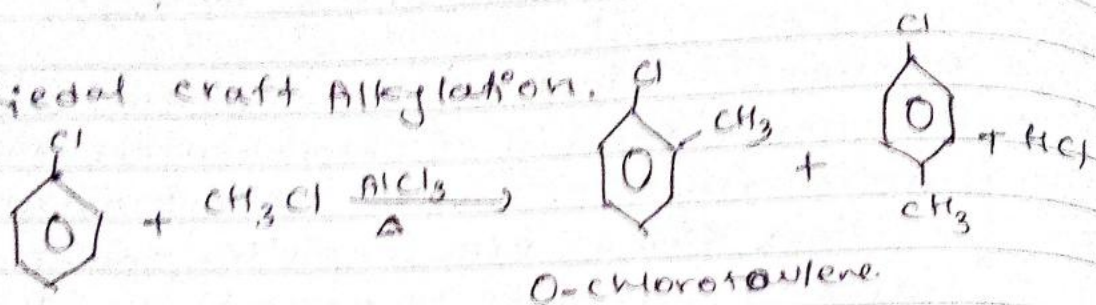
b) Sulphonation.



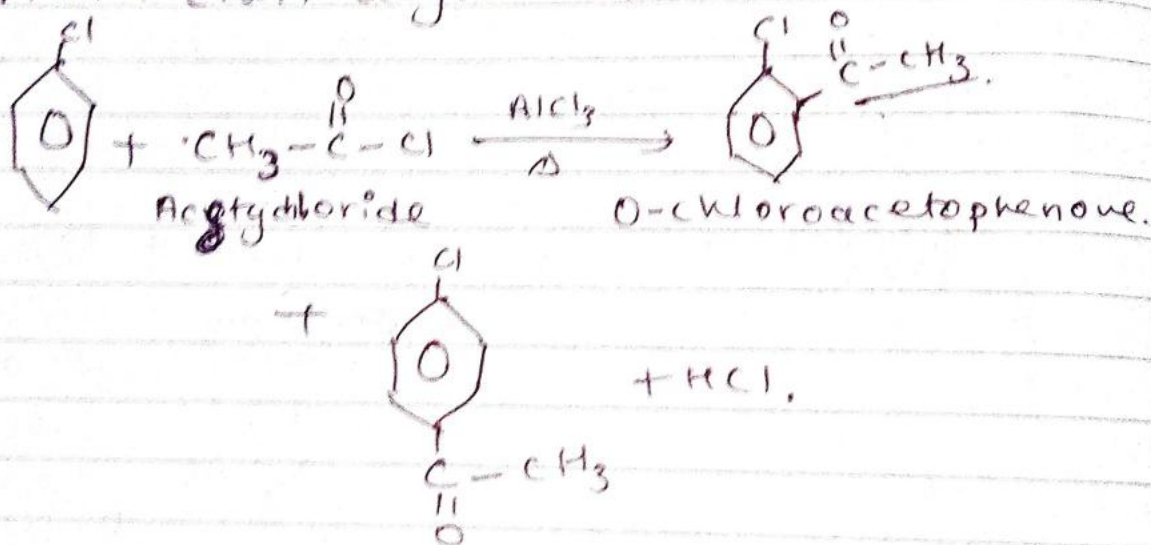
c) Nitration:



d) Friedel craft Alkylation.

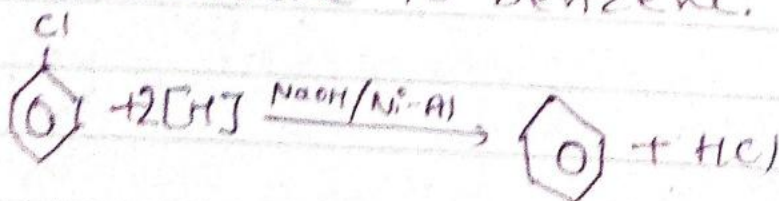


e) Friedel craft acylation.



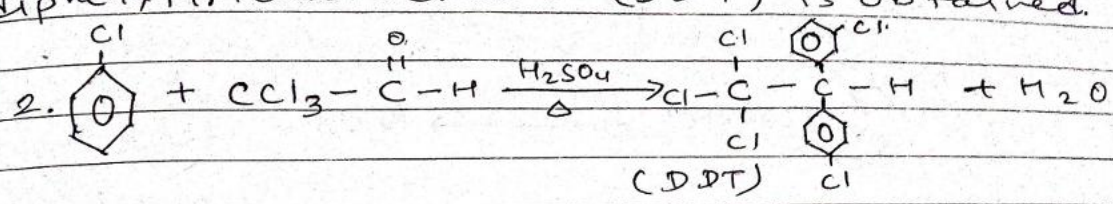
② Reduction

→ In alkaline medium, Ni-Al alloy reduces chlorobenzene to benzene.



③ Rxn with chloral

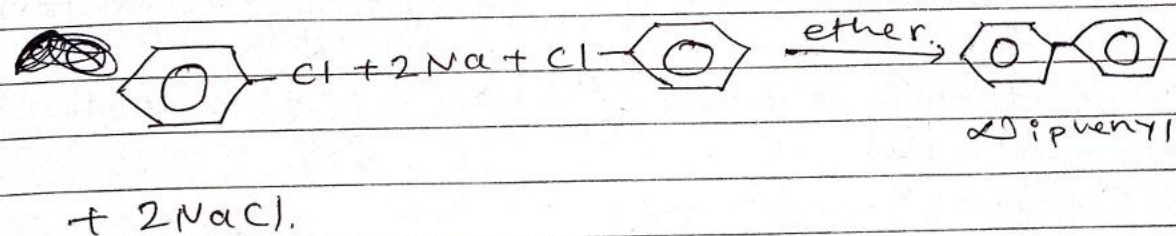
→ Chlorobenzene when heated with chloral in presence of conc. sulphuric acid, dichloro diphenyl trichloroethane (DDT) is obtained.



→ DDT is used as insecticides.

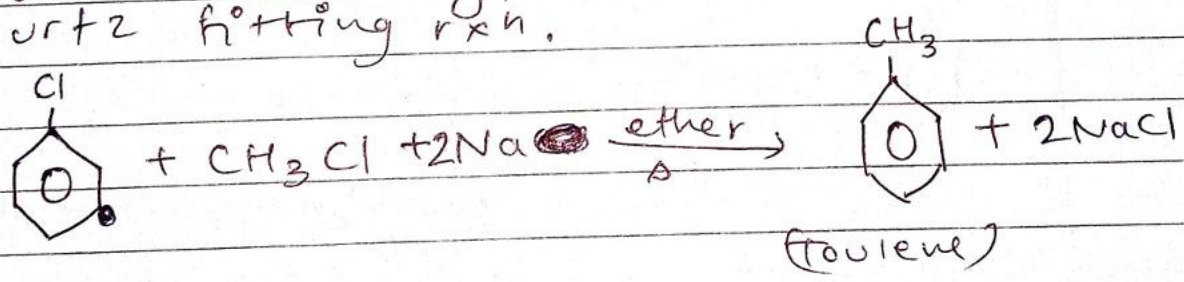
④ Rxn with sodium (Fitting rxn)

→ Chlorobenzene when heated with sodium metal in presence of ether, diphenyl is obtained.



⑤ Wurtz Fitting rxn.

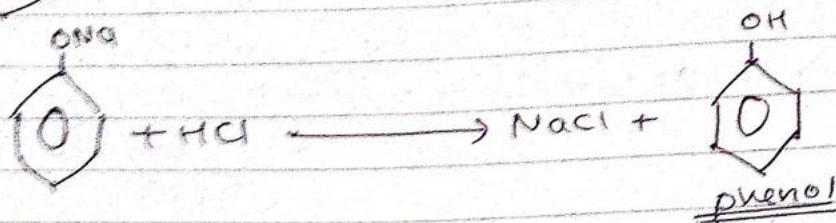
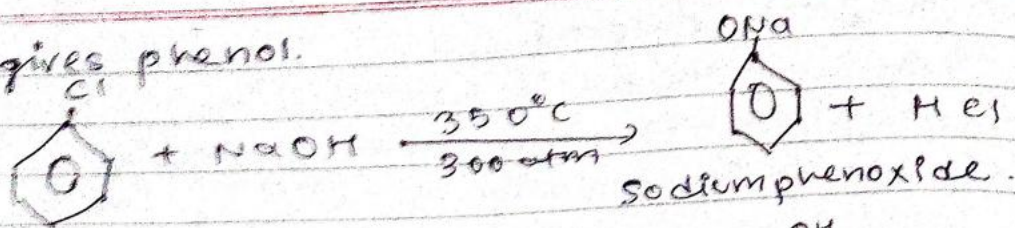
→ When chlorobenzene is heated with haloalkane in presence of ether, alkyl benzene is obtained. This type of rxn is called as wurtz fitting rxn.



⑥ Rxn with NaOH

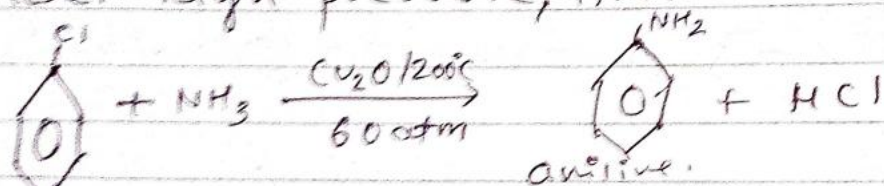
→ Chlorobenzene when strongly heated with NaOH, under high pressure, sodium phenoxide is obtained which on acidification,

gives phenol.

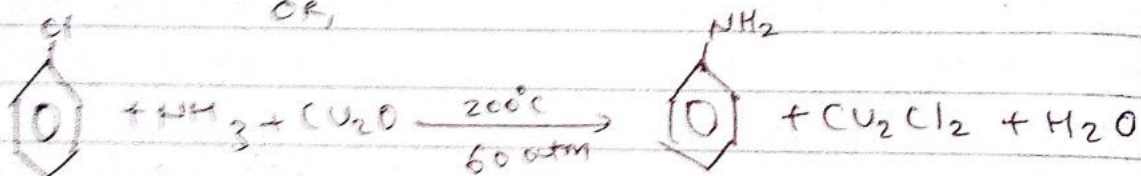


④ Rxn with Ammonia (Ammonolysis).

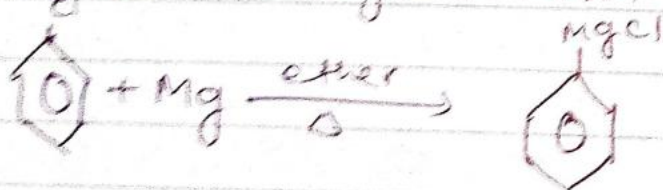
→ Chlorobenzene when strongly heated with ammonia in presence of cuprous oxide under high pressure, Aniline is formed.



OR,



⑤ Rxn with magnesium.



phenylmagnesium chloride.

⑥ Rxn with cuprous cyanide.

